WMED 2026 CONFERENCE PRESENTATION

Validation Proportion Matters: Generalization Stability of YOLO-Based Wafer Defect Detection Under Limited Data

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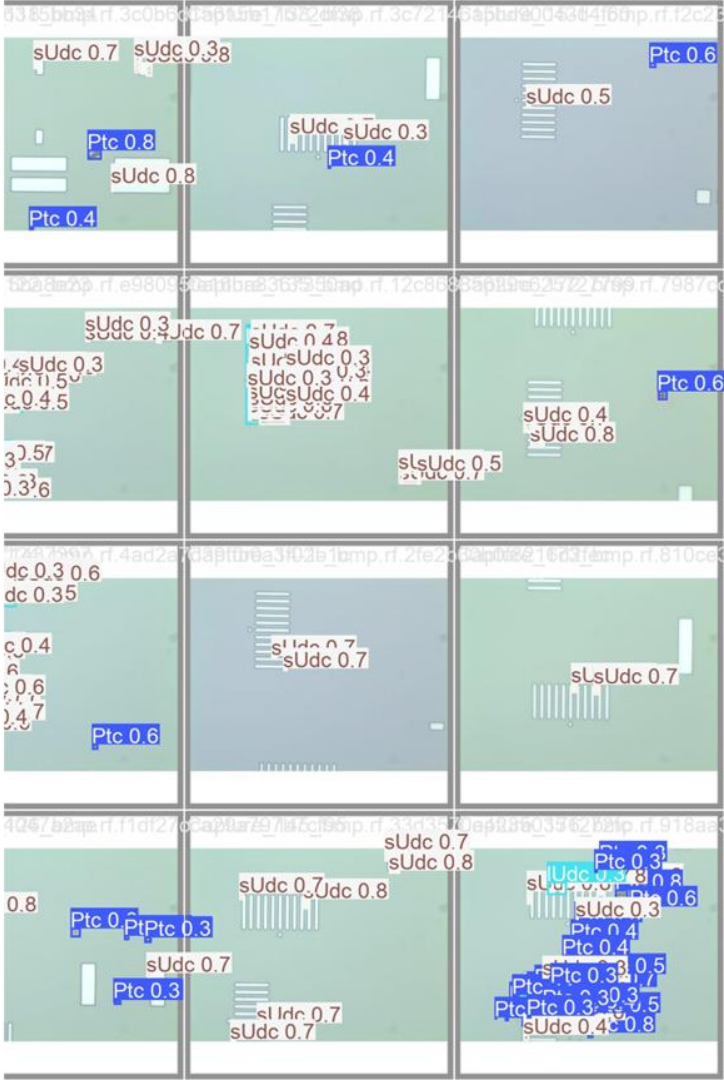

Presentation Overview

This presentation covers a controlled experimental study examining how dataset partitioning strategy — specifically validation proportion — affects the generalization stability of YOLO-based defect detection models under limited semiconductor data conditions.

01

Motivation & Problem

Why validation proportion is under-explored in semiconductor inspection

02

Experimental Setup

Dataset, defect classes, partition strategies, and model architectures

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Methodology

Training procedure, augmentation pipeline, and evaluation protocol

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Performance metrics, convergence behavior, and class-level findings

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Discussion & Conclusion

Key takeaways and directions for future work

MOTIVATION

The Challenge of Defect Inspection in Semiconductor Manufacturing

Surface defect inspection is a critical gating step in semiconductor process control. Defects such as undercut formation and particulate contamination can degrade structural integrity, distort lithographic patterns, and reduce device yield. Traditional rule-based systems are highly sensitive to illumination variation and surface texture complexity, limiting their robustness in advanced fabrication environments.

Undercut Defects

Caused by isotropic etching — lateral material removal beneath the photoresist mask compromises pattern fidelity

Particulate Contamination

Foreign particles from airborne debris or process residues interfere with lithography, etching, and thin-film deposition

Limited Labeled Data

Acquisition constraints and manual labeling costs keep semiconductor inspection datasets small, complicating deep learning deployment

RESEARCH GAP

Why Validation Proportion Has Been Overlooked

The Common Assumption

Practitioners commonly assume that maximizing training data size is the primary lever for improving model performance. Under limited data conditions, increasing the training proportion to 80% seems intuitive — more examples should mean better generalization.

The Overlooked Risk

Insufficient validation sampling produces a noisy supervisory signal. When the validation set is too small, early stopping triggers on statistical artifacts rather than true generalization degradation, leading to sub-optimal weight selection.

Our Research Question

For small semiconductor datasets, does increasing training proportion to 80% actually improve detection performance — or does maintaining a larger validation set (20%) yield more stable generalization?

Despite its practical importance, the influence of **validation proportion on generalization stability** remains largely unexplored in semiconductor defect detection literature.

EXPERIMENTAL SETUP

Dataset: 300-Image Wafer Microscopy Collection

The dataset consists of **300 high-resolution wafer microscopy images** captured at 5× magnification under cleanroom illumination. All images were annotated with bounding boxes across three defect classes using Roboflow. Class distribution is highly imbalanced, with particles (~1,200 instances) and small-area undercut (~2,100 instances) vastly outnumbering large-area undercut (~150 instances).

sUdc — Small-Area Undercut

~2,100 instances · 2–10 instances/image · Peak: 37. Localized lateral etching beneath the photoresist layer due to isotropic etch processes.

lUdc — Large-Area Undercut

~150 instances · 1 instance/image · Peak: 3. Coalescence of multiple sUdc regions into a continuous extended lateral etch zone.

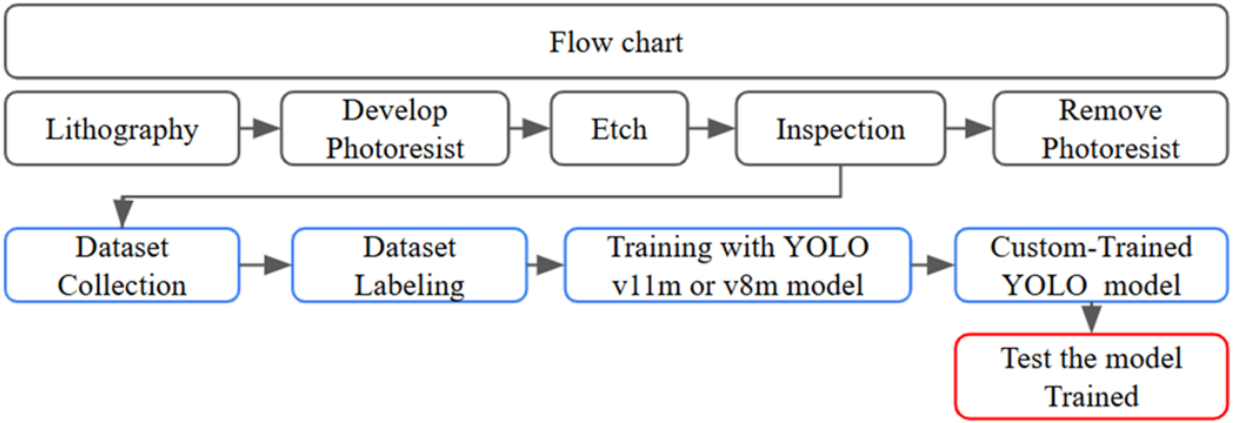
Ptc — Particles

~1,200 instances · 2–25 instances/image · Peak: 29. Foreign particulate contamination from airborne debris, process residues, or handling artifacts.

EXPERIMENTAL SETUP

Three Dataset Partition Strategies

Three train/validation/test split configurations were designed to systematically analyze the trade-off between training capacity and validation reliability. Data splitting was performed on the original 300 images **prior to augmentation** to prevent data leakage. Validation and test sets contain strictly original, non-augmented images.



A 3× augmentation factor was applied exclusively to training partitions. The 70/20/10 strategy allocates 60 original images to validation, providing the richest supervisory signal for the rare IUdc class and early stopping decisions.

EXPERIMENTAL SETUP

Model Architectures: YOLOv8m & YOLOv11m

YOLOv8m

A mature medium-scale single-stage detector formulating detection as a regression problem. Features a decoupled head design separating classification and localization. Pre-trained on COCO (80 classes); classification head replaced with a 3-class linear layer (sUdc, lUdc, Ptc) via transfer learning.

YOLOv11m

An architecturally refined successor incorporating enhanced feature fusion, improved small-object localization, and greater parameter efficiency. Same decoupled head modification applied. YOLOv11m's deeper hierarchical feature extraction benefits from longer training under the 70/20/10 split (converged at epoch 93 vs. YOLOv8m's epoch 28).

Optimizer

AdamW · lr = 0.0003 · weight decay = 0.0005

Training

Max 200 epochs · Early stopping: 50-epoch patience

Input

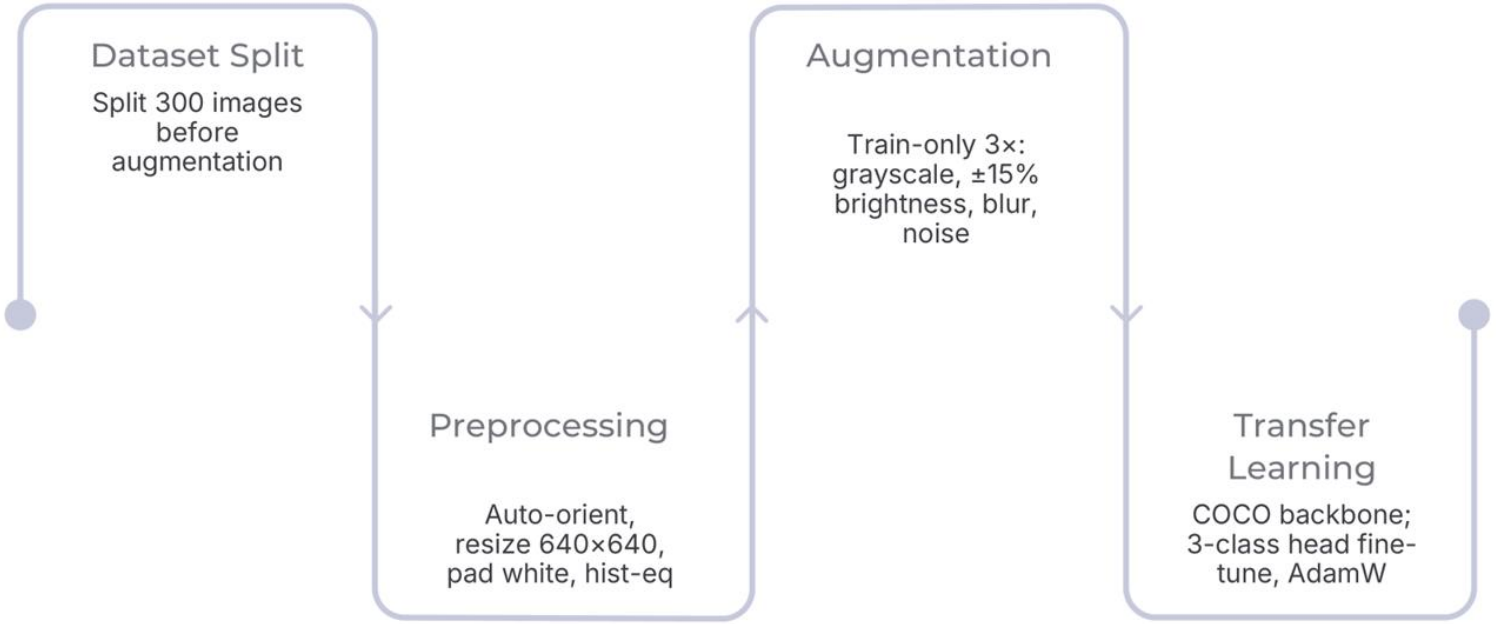
640×640 px · Batch size: 12

Hardware

NVIDIA RTX 4060 · cuDNN · Ultralytics framework

METHODOLOGY

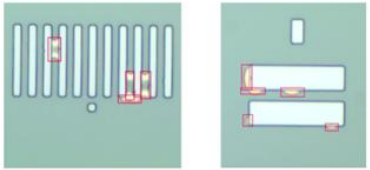
Training Pipeline & Data Augmentation



Preprocessing included global histogram equalization applied to the training set to highlight undercut boundaries. Augmentation was strictly limited to the training partition, ensuring validation and test sets reflect real-world, unmodified wafer image conditions. This design choice prevents the model from benefiting artificially from augmented validation samples.

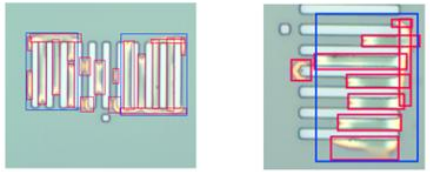
METHODOLOGY

Defect Class Definitions & Labeling



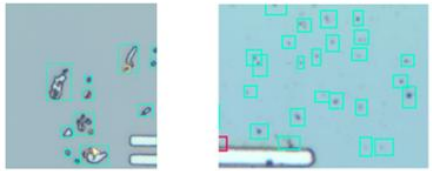
Small-Area Undercut (sUdc)

Localized lateral etching beneath the photoresist layer. Results from isotropic etch processes where material removal occurs in both vertical and lateral directions, causing unintended sidewall recession beneath the PR mask. Most frequent class with up to 37 instances per image.



Large-Area Undercut (lUdc)

Coalescence of multiple sUdc regions into a continuous, extended lateral etch zone. May spatially overlap with or fully encompass sUdc regions. Caused by prolonged etch duration, excessive chemical activity, or inadequate mask adhesion. Rarest class (~150 total instances).



Particles (Ptc)

Foreign particulate contamination from airborne debris, process residues, equipment wear, or handling artifacts. Particles interfere with photolithography, etching uniformity, and thin-film deposition, potentially causing pattern distortion or device failure. Critical for yield control.

METHODOLOGY

Why 20% Validation? The Statistical Rationale

The Instance-Level Argument

A 10% validation split yields only 30 images — a statistically thin supervisory signal for rare defects like IUdc (~150 total instances). Many validation batches may contain zero IUdc examples, creating an unreliable loss landscape for early stopping.

Allocating 20% (60 images) to validation exposes the model to **600+ total defect instances** during evaluation each epoch, providing a far more representative sample of class diversity.

Impact on Convergence

Under the 80/10/10 split, YOLOv11m's validation classification loss spiked to **1.002** — a clear indicator that the 30-image validation set could not reliably distinguish true generalization failure from localized sampling noise.

Under 70/20/10, the same metric stabilized at **0.960**, enabling early stopping to trigger on genuine generalization signals rather than statistical artifacts. This allowed YOLOv11m to train until epoch 93, nearly triple the duration of the 70/15/15 split.

RESULTS

Overall Detection Performance: All Models & Splits

Performance was evaluated using Precision, Recall, mAP50, and mAP50–95 following the COCO evaluation protocol. The table below summarizes results across all six experimental configurations. **YOLOv11m with the 70/20/10 split consistently achieves the highest mAP50 and Recall.**

Model	Split	mAP50	mAP50–95	Precision	Recall
YOLOv8m	70/15/15	0.801	0.437	0.772	0.765
YOLOv8m	70/20/10	0.805	0.464	0.791	0.784
YOLOv8m	80/10/10	0.782	0.456	0.742	0.731
YOLOv11m	70/15/15	0.804	0.452	0.785	0.795
YOLOv11m	**70/20/10**	**0.819**	**0.460**	**0.781**	**0.810**
YOLOv11m	80/10/10	0.792	0.447	0.755	0.772

Notably, the 80/10/10 split degraded performance for both architectures relative to the 70/20/10 configuration, confirming that increased training volume cannot compensate for an insufficient validation signal.

RESULTS

Key Comparison: 70/20/10 vs. 80/10/10 on YOLOv11m

0.821

mAP50 — 70/20/10

Best overall detection accuracy achieved under the larger validation split

0.791

mAP50 — 80/10/10

Reduced performance despite 10% more training data

0.831

Recall — 70/20/10

Higher recall critical for missed-defect reduction in fab environments

0.960

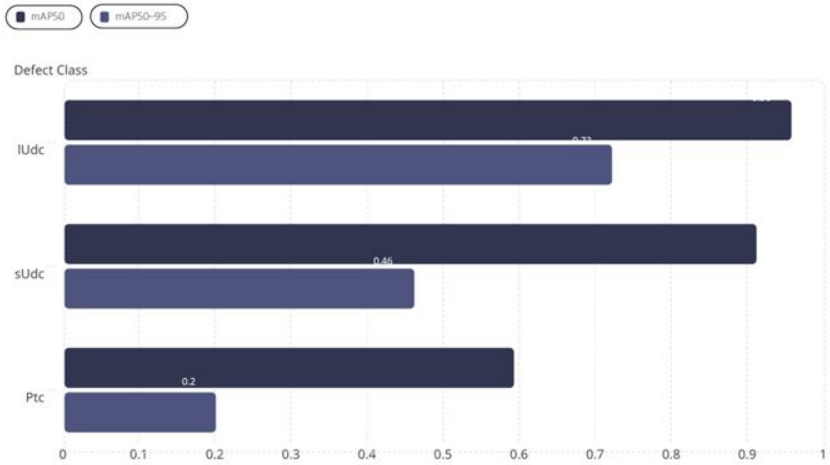
Val Cls Loss — 70/20/10

Stable convergence signal enabling reliable early stopping at epoch 93

The 80/10/10 split exhibited a validation classification loss spike to **1.002**, indicating noisy convergence monitoring caused by an insufficiently small 30-image validation set.

RESULTS

Class-Level Detection Performance



Structural Defects Excel

Both undercut classes achieve strong detection performance. lUdc reaches an impressive **mAP50 of 0.956** and mAP50–95 of 0.720, while sUdc achieves mAP50 ≈ 0.91 — demonstrating that the model reliably localizes structural etching defects at multiple IoU thresholds.

Particle Detection Gap

Ptc achieves mAP50 = 0.591 but only mAP50–95 = 0.199, revealing a **generalization gap**. The model identifies particle presence (high recall trigger) but struggles with pixel-level boundary precision — likely due to small size, variable morphology, and intra-class visual diversity.

RESULTS

Convergence Behavior & Learning Rate Analysis

A progressive learning rate decay strategy was adopted, with final lr/pg0 values ranging between **1.0×10^{-4} and 1.9×10^{-4}** across all configurations. The AdamW optimizer decoupled weight decay from gradient updates, improving regularization stability across all splits.

YOLOv11m — 70/20/10 Split

Converged at **epoch 93**. The extended training duration allowed the deeper architecture to fully exploit its hierarchical feature extraction capability, yielding the highest mAP50 = 0.821 and val cls loss = 0.960.

YOLOv8m — 70/20/10 Split

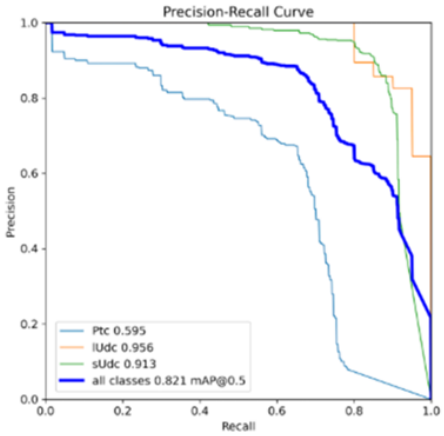
Converged earlier at **epoch 28**, reflecting the shallower architecture's faster but less thorough feature learning. Achieved competitive mAP50 = 0.805 under the same split.

80/10/10 Splits — Both Architectures

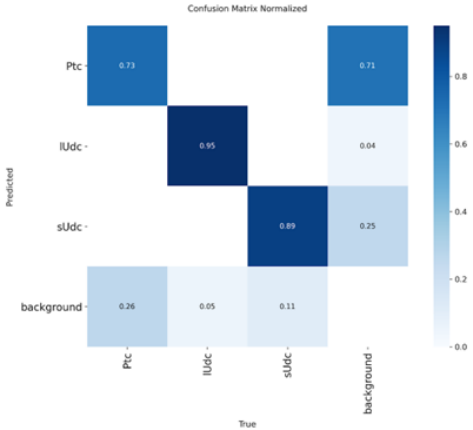
Validation cls loss spiked to **1.002** for YOLOv11m, indicating that the 30-image validation set produced a noisy supervisory signal insufficient for reliable early stopping and model selection.

RESULTS

Precision–Recall Curves & Qualitative Predictions



PR Curve — YOLOv11m 70/20/10: Area under the PR curve confirms strong class-level separation for lUdc and sUdc. The Ptc curve shows a precision-heavy but area-limited profile, consistent with the mAP50–95 gap.



High Accuracy: Excellent classification for lUdc (95%) and sUdc (89%), showing precise localization of structural defects.
Minimal Class Confusion: Very low misclassification between different defect types, confirming strong feature extraction.
Background Sensitivity: Primary challenge remains in background texture, where high-frequency noise leads to occasional false positives for Ptc.
Key Insight: Model performance is robust for critical defects but sensitive to small-object detection under limited data constraints.

DISCUSSION

The Core Trade-Off: Training Volume vs. Validation Reliability

The central intellectual contribution of this study is a clear articulation of the trade-off between training volume and validation reliability in small-scale industrial datasets. The results demonstrate that a larger validation set forces the model to prove generalization across more unseen data at every epoch, significantly reducing the risk of *stabilize convergence* — a common failure mode in small datasets where early stopping triggers on a validation artifact rather than a true signal.

More Training Data (80/10/10)

30-image validation set → noisy supervisory signal → unreliable early stopping → sub-optimal weight selection → **lower mAP50 and recall**

Balanced Validation (70/20/10)

60-image validation set → stable convergence signal → reliable early stopping at epoch 93 → optimal weight selection → **highest mAP50 and recall**

DISCUSSION

Limitations & Areas for Future Refinement

Particle Localization Gap

The mAP50–mAP50–95 gap of ≈0.36–0.38 indicates that high-IoU bounding box regression for small particles remains limited. Moving to 10× magnification and advanced loss formulations (e.g., Generalized Focal Loss) may improve boundary precision.

Dataset Scale

The 300-image dataset, while carefully constructed, limits statistical generalizability. Larger, more diverse datasets spanning multiple fab tools and process nodes will be required for industrial deployment readiness.

Single-Seed Evaluation

Current benchmarks do not include multi-seed cross-validation. Future work will standardize multi-seed protocols to quantify the statistical robustness of the observed performance gains and reduce partition-induced variance.

Horizontal Bounding Boxes

Standard axis-aligned detection cannot capture directional etch bias. Oriented bounding boxes (OBB) will be explored to enable directional etch analysis and tool-induced non-uniformity diagnostics.

FUTURE WORK

Roadmap for Bridging Academic Models to Fab-Floor Requirements

1. **Near Term**

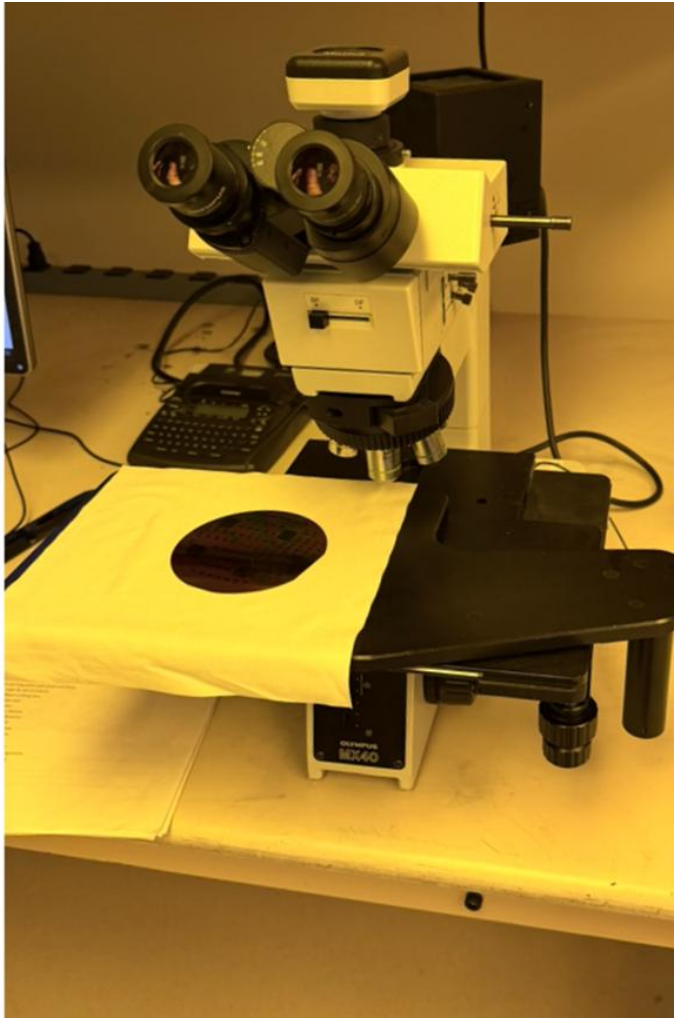
   Multi-seed cross-validation to quantify statistical robustness; weighted loss formulations targeting critical false negatives in high-stakes process steps

2. **Medium Term**

   Oriented bounding box (OBB) detection for directional etch analysis; integration of Generalized Focal Loss for improved high-IoU localization

3. **Long Term**

   Larger, multi-tool, multi-node datasets; higher magnification (10×) imaging for particle metrology; deployment-grade real-time inspection pipeline



CONCLUSION

Key Takeaways

1 Validation Proportion Outweighs Training Volume

Under limited semiconductor data conditions, maintaining a 20% validation set yields more stable generalization and higher mAP50 than increasing training proportion to 80%. This finding challenges the common assumption that more training data is always better.

2 YOLOv11m + 70/20/10 Achieves Best Performance

This configuration achieved mAP50 = 0.819, Recall = 0.810 — the highest recorded across all six experimental conditions. The 60-image validation set provided the supervisory richness needed for stable convergence through epoch 93.

3 Structural Defects Are Well-Detected; Particles Need Refinement

lUdc (mAP50 = 0.956) and sUdc (mAP50 ≈ 0.91) achieve strong localization. Particulate contamination (mAP50 = 0.591, mAP50–95 = 0.199) requires architectural and imaging improvements for high-precision metrology.

4 A Practical Guideline for Limited-Data Industrial Benchmarks

Dataset partition strategy should be treated as a first-class hyperparameter in semiconductor defect detection pipelines. A minimum 20% validation allocation is recommended when total dataset size is below ~500 images.

Q&A

LinkedIn

GitHub

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